Program overview

12-Sep-2023 13:19

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master TM 2023	**Master Technical Medicine 2023**						
TM Obligatory Courses	**KT Obligatory Courses**						
TM10002	Python Programming	2,5					
TM10003	Brok	1,5					
TM10004	Medical Skills in Daily Practice	3					
TM10006	Patient Cases	7,5					
TM10007	Machine Learning	2,5					
TM10008	Patient Journey	1					
TM10009	Skills in Acute Setting	2					
TM Track Imaging & Intervention							
TM11001	Advanced Image Acquisition	5					
TM11002	Molecular Imaging and Therapy	5					
TM11003	Radiation Protection	5					
TM11004	Radiation Therapy	5					
TM11005	Advanced Image Processing	5					
TM11006	Image Guided Interventions	5					
TM11007	Biomaterials and Issue Biomechanics	5					
TM11008	Computer Assisted Reconstructive Surgery	5					
TM Track Sensing & Stimulation							
TM12001	Advanced Signal Acquisition	5					
TM12002	Sensing of Neurophysiological Signals	5					
TM12003	Electrostimulation of Neurophysiological systems	5					
TM12004	Drug Sensing and Delivery	5					
TM12005	Advanced Signal Processing	5					
TM12006	Extramural Sensing and Virtual Stimulation	5					
TM12007	Sensing and Stimulation of Circulation and Ventilation: Acute and Chronic Care	10					
TM Second Year							
TM20005	TM Clinical Internship 1	14					
TM20006	TM Clinical Internship 2	14					
TM20007	TM Clinical Internship 3	14					
TM20008	TM Clinical Internship 4	14					
TM20009	Medical Technology in Clinical Practice	4					
TM Third Year							
TM30001	TM Practical Internship	15					
TM30002	TM Thesis Feasibility Study	15					
TM30003	TM Literature Study	10					
TM30004	TM MSc Thesis	35					
15 EC Elective courses							

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

Master TM 2023	
Contact for students	master-tm@tudelft.nl
In association with the University of	Leiden Universty (LUMC) Delft University of Technology (3mE) Erasmus University Rotterdam (Erasmus MC)
EC Program	180
Program Start (Study Year)	September (including students from other universities) February (excluding students from other universities)
Prerequisites	Degree in Clinical Technology (Bachelor)
Introduction 2	Information about courses of year 2 will be published in june 2018
Introduction 3	Information about courses of year 3 will be published in june 2019

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

TM Obligatory Courses

TM10002	**Python Programming**	2.5
Responsible Instructor	Dr.ir. T. van Walsum	
Contact Hours / Week x/x/x/x	Week 1: 1h HC / 6h WC / 5h ZS Week 2: 1h HC / 3h WC / 3h ZS / 5h GO Week 3: 12h GO Week 4: 1h WC / 2h ZS / 9h GO Week 5: 2h ZS / 8h GO / 2h Exam	
Education Period	3A	
Start Education	3A	
Exam Period	3A 4A	
Course Language	English	
Course Contents	The students have acquired some programming experience with Matlab in the BSc programme. This course will focus on learning how to use the Python programming language to solve coding problems. The problem-oriented teaching approach starts from a concrete problem and builds up knowledge in python programming up to the level required to solve the problem. By working as a team on a programming project, the students will learn how to decompose the programming problem in blocks and how to define adequate interfaces between the blocks. The students will be familiarized with the tools for programming in teams and building, debugging and testing their code. Topics of the course are: - Introduction to python programming. - Scientific programming. - Source code version control.	
Study Goals	Students are able to: 1. Write scripts using python. 2. Implement correctly working code using python. 3. Adequately use version control and develop testing routines and procedures. 4. Use scientific computing python packages	
Study Goals continuation	Contribution to learning outcomes A1, A4, C2, D1, D6	
Education Method	Mandatory - Group project - Video lectures Advised - Exercises with video lectures - Self-study	
Literature and Study Materials	DVCS tutorial / Python tutorial / Python scientific packages tutorial	
Assessment	The final grade consists of: - Result of individual examination - Result of group assignment (programming project) Both have a weight of 0.5 and both should be 5.0 or higher. The individual exam may consist of a written examination and/or programming exercises. In case of unforeseen circumstances or measures resulting from COVID-19, the individual written examination may be replaced with an on-line examination (individually graded, exam individually or in small groups).	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	Theo van Walsum: t.vanwalsum@erasmusmc.nl Hakim Achterberg: h.achterberg@erasmusmc.nl	

TM10003	Brok	1.5
Responsible Instructor	Ir. A.G. van Tilborg	
Contact Hours / Week x/x/x/x	more information in course guide (blokboek)	
Education Period	None (Self Study)	
Start Education	1A 3A	
Exam Period	Exam by appointment	
Course Language	Dutch English	
Course Contents	The students will be familiarized with the laws and regulations on medical research with patients or patient material. Since 2006, all clinical researchers are obliged to follow the BROK course, to make sure that all researchers know and abide the law and regulations when performing research with patients.	
Study Goals	The student: has in-depth knowledge of rules and regulations on patient-based research and is able to apply the rules and regulation on his own research. For the detailed course objectives see http://emwo.nl/wp-content/uploads/2022/09/Testmatrix-BROK-and-GCP-WMO-exams.pdf	
Study Goals continuation	Contribution to learning outcomes: D3-5, E4, F2,8	
Education Method	- Self-assessment - Weblectures, on line reading material, animations - Forum, group discussions and polls - Lectures Attending the (digital) CSB (centrum specifieke bijeenkomst) is obligatory. The eBROK contains ± 15 hours of interactive learning material, practice material, videos and animations. Divided into 5 basic modules, 4 in-depth modules and a practice module for practice. With the help of experts from the field, all learning material has been carefully designed.The eBROK consists of six compulsory modules and four in-depth modules. This is a self-study course and can be taken throughout the academic year	
Literature and Study Materials	The course material will be provided via internet.	
Assessment	The final grade consists of: - Written examination Examination can be held on location (Erasmus MC) or online via proctoring exam.	
Enrolment / Application	This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The course is offered online by the NFU. https://nfu-ebrok.nl/?lang=en	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen.	
Contact	brok@erasmusmc.nl or A.G.vanTilborg@tudelft.nl	

TM10004	Medical Skills in Daily Practice	3
Responsible Instructor	Dr. J. Damen	
Contact Hours / Week x/x/x/x	more information in course guide (blokboek)	
Education Period	1A 1B 2A 4A 4B	
Start Education	1A 1B 2A 4A 4B	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	This course focuses on the training of basic medical technical skills and professional skills to meet the requirements for the BIG-registration and develop competency in the medical profession. Professional skills: - Sterility and rules of conduct on the OK Medical technical skills: - Suturing - Bladder catheterization - Vena punction and IV cannulation The technical modules are structured as follows 1.E-learning module 2.Practising of skill 3.Test Key: For admission to the test of the modules, the student must have followed the e-learning modules and have practiced each technical module at three practice sessions.	
Course Contents Continuation	Contribution to learning outcomes: A3, A5, A6, B2, B5, D1,F4,F7	
Study Goals	Students are able to: 1. Perform pre-operative hand disinfection 2. Put on a sterile surgical gown 3. Put on sterile gloves according to the open method 4. Suture and knot 5. Perform a punction (venous punction, peripheral venous access) 6. Perform a catheterization for a male and a female patient 7. Reflect on ones own learning capabilities, ask feedback and set future learning goals	
Education Method	Mandatory - Practical training in simulation and real-life setting - E-learning	
Literature and Study Materials	Study material will be provided on Brightspace	
Assessment	The final grade consists of: - Demonstration of medical skills (pass or fail) - Demonstration of professional skills (pass or fail) Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.6 states the following: The result of a professions-act examination is valid for a two year period, until start of the clinical internships.. Participation in the Internship courses [TM20001, TM20002, TM20003, and TM20004] depends on the condition of having successfully completed TM10004, TM10006, TM10008, and TM10009 within 24 months prior to enrolment in TM20001. In the event the student does not meet this criterion, (s)he needs to resit that particular course. Upon successful completion of the resit(s), the student is eligible to enrol in TM20001 (and TM20002, TM20003, and TM20004].	
Permitted Materials during Tests	none	
Enrolment / Application	Every 2 weeks a new group starts, both in semester 1 and 2. For each group, the course is scheduled in 5 consecutive weeks. The student has to enroll her/himself in a specific group via Brightspace after registering in MyTUDelft. Note that this course can be taken in the same semester as TM10009, but NOT in the same five week period. This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	If the student has passed the test for a medical skill, the student will receive a certificate with a validity of 2 years. All skills must be mastered before entering the clinical internships. The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. To this end he/she acquires a (temporary) BIG registration of 9 manual medical skills some of which are trained in this course.	

Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	onderwijs.medicalskills1@erasmusmc.nl

TM10006	Patient Cases	7.5
Responsible Instructor	Dr. M.S. Arbous	
Module Manager	Drs. N.F. Klijn	
Module Manager	Prof.dr. J.I. Rotmans	
Module Manager	Dr. I.A. van Rossum	
Module Manager	Drs. J.D. Rintjap	
Module Manager	Dr. J.M.J. Boogers	
Contact Hours / Week x/x/x/x	2-4 hours per week, 6 10 hours of self-study per week	
Education Period	1B 2A 2B 4A 4B	
Start Education	1B 4A	
Exam Period	Different, to be announced	
Course Language	Dutch (on request English)	
Course Contents	The patient is the centre of the medical universe around which all our work revolves and towards which all our efforts tend William Murphy, 1934 Nobel Prize winner. The Technical Medicine Professional, like medical doctors, should approach patients from multiple perspectives ranging from large-scale prevention to individual patient care. In collaboration, both professionals should be able to conceptualize care around patients and use technology to improve the diagnostic and treatment process. In this process, the medical interview and physical examination are specific skills which play an essential role in the communication between (potential) patients and healthcare providers. It is used to guide patients from complaint to diagnosis and treatment in a structured form, keeping in mind the patient profile (background, age, expectations, desires, etc). As such, it forms the basis of treatment relationships in most healthcare contexts and determines where technology can be used to improve patient outcomes. This case-based course addresses communication skills in a healthcare context, the diagnostic process, patient treatment and follow-up. Students will learn where (the development of) technology can be helpful and how it is integrated in modern patient care. The first cases will focus on basic communication skills and physical examination, the latter cases will focus on integration of technology into patient-care. Topics include: - The medical interview - Physical examination - Communication skills in a healthcare context - The diagnostic process - From diagnosis to treatment - Conceptualizing medical problems into technological solutions	
Study Goals	The students are able to: 1. Perform the medical interview 2. Perform a full physical and neurological examination 3. Understand the different steps within the diagnosis, treatment and follow-up process 4. Explain the (potential) (mis)use of technology within the diagnosis, treatment and follow-up process 5. Identify and frame relevant technological problems in medical practice 6. Conceptualize solutions for medical-technological problems	
Education Method	Working groups to solve cases, self-study, practical training, coach meetings, interaction with real and simulation patients and lectures.	
Literature and Study Materials	- Grundmeijer HGLM, Reenders K, Rutten GEHM. Het geneeskundig proces (3e of 4e druk). Maarsen, 2009 - Henry MM, Thompson JN. Clnical Surgery (3e druk). Elsevier Health Service 2012. - Hijdra A, Koudsstaal PJ, Roos RAC. Neurologie (6e druk). Springer Media B.V. 2015 (inclusief filmmateriaal op StudieCloud) - De Jongh TOH, de Vries H, Grundmeijer HGLM. Diagnostiek van alledaagse klachten (3e druk) Houten/Diegem: Bohn Stafleu Van Loghum, 2011 - Van der Meer J , van der Meer JWM. Anamnese en lichamelijk onderzoek (5e of 6e druk). Elsevier Gezondheidszorg, Maarsen, 2008 - Kumar, P, Clark, M. Clinical Medicine (8e editie), 2012-2013 Elsevier Healthe Sciences - Lloyd M, Bor R. Communication Skills for Medicine, (3rd ed.). Edinburgh: Churchill Livingstone, 2009 - Thijs, LG, Delooz, HH, Goris, RJA, Schers, HJ. Acute Geneeskunde. Elsevier gezondheidszorg, Maarssen, zesde herziende druk, 2005	
Assessment	All cases are assessed in different ways, depending on the case. There will be discussions, presentations, (simulated) patient interactions and clinical visits. Every case has to be passed to pass the whole course. If one case is not passed, there is an opportunity to pass an extra case. If you fail on more than one case, you will have to redo the course. To be submitted in each case: Case 1: a medical specialist letter to the general practitioner concerning your case, a medical technical problem analysis Case 2: a medical specialist letter to the general practitioner concerning your case. To be performed and passed: a medical interview with and physical examination of a (simulation) patient with dyspnea Case 3: a medical technical problem analysis with a possible medical technical solution Case 4: a medical technical problem analysis with a possible medical technical solution. To be performed: neurological physical examination Case 5: a medical technical problem analysis with a possible medical technical solution (thorough) Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.6 states the following: The result of a professions-act examination is valid for a two year period, until start of the clinical internships.. Participation in the Internship courses [TM20001, TM20002, TM20003, and TM20004] depends on the condition of having successfully completed TM10004, TM10006, TM10008, and TM10009 within 24 months prior to enrolment in TM20001. In the event the student does not meet this criterion, (s)he needs to resit that particular course. Upon successful completion of the resit(s), the student is eligible to enrol in TM20001 (and TM20002, TM20003, and TM20004].	
Permitted Materials during Tests	During practical sessions no learning materials are permitted	
Enrolment / Application	This course is only open to fulltime TM-students.	

	To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. To this end he/she acquires a (temporary) BIG registration of 9 manual medical skills which are trained in this course. In addition (inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	M.S. Arbous, M.S.Arbous@lumc.nl

TM10007	**Machine Learning**	**2.5**
Responsible Instructor	Dr. J.F. Veenland	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4B	
Course Language	English	
Expected prior knowledge	The student - has succesfully completed the Python course, or demonstrates proficency in Python and - has basic knowledge of machine learning (level of KT3601)	
Course Contents	Datascience and Machine Learning (Deep Learning) techniques can be applied in all fields in medicine were data, signals and images are involved: from HIX-data, retina OCT images, prostate MRI to ECGs, to genomics and biomics data. This course familiarises students with the basic concepts and methods of Machine Learning. The course will cover several topics such as feature selection methods, dimensionality reduction techniques, classifiers, deep learning, performance metrics, generalization, overfitting and regularization. Examples and assignments will be primarily from the medical field.	
Study Goals	- The student understands the basic machine learning concepts and methods. - The student is able to select and implement ML methods in Python to well-defined problems and to interpret and evaluate the results - The student is able to justify the choices in the chosen ML-strategy - The student is able to identify and explain the pitfalls in ML strategies, and the advantages and limitations of several ML related methods. - The student is able to reflect on the added value of machine learning applications in the medical domain	
Education Method	Advised - Lectures - Video-clips - Working groups - Reading material - Self study - Programming - Exercises Mandatory - Assignment (group)	
Prerequisites	Python course (TM10002) or demonstrate proficency in Python	
Assessment	The final grade consists of: - Written examination (50%) - Group assignment (50%) Both should be 5.0 or higher. Weighted average should be 5.8 or higher.	
Enrolment / Application	This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

TM10008	Patient Journey	1
Responsible Instructor	Dr. R.W.C. Scherptong	
Contact Hours / Week x/x/x/x	more information in course guide (blokboek)	
Education Period	2B	
Start Education	2B	
Exam Period	2B	
Course Language	Dutch (on request English)	
Course Contents	As a medical technical professional it is important to understand the perspectives of all stake-holders within care-processes. Hospital care frequently involves a large number of (para)medical professionals, supportive staff and multiple departments. Organization of care-processes can be counter-intuitive, which makes it difficult for patients to find their way through the complex network of care-providers. The addition of technology within this process addresses medical challenges, however may further increase complexity from the patient perspective. To work as a medical technical specialists within the hospital setting requires general knowledge about hospital organization and specific knowledge about the organization of care-tracks surrounding patients and understanding of patient-journeys to predict or evaluate quality issues of novel approaches to patient care. In this course, students will learn the basics behind patient journeys. Knowledge about these subjects is regarded essential before entering the practical internships.	
Study Goals	1. To understand the function of patient journeys within hospitals. 2. To be able to define, outline and connect the relevant actors within patient journeys 3. To be able to sketch a patient journey, through interviews with professionals and patients	
Study Goals continuation	A 1-4; B2; C1-2; D 1-5; F 1-3, 8	
Education Method	Online and on campus when appropriate	
Assessment	The study product of this course is a group assignment. In groups of maximum 4 students a Patient Journey will be created, presented on a poster and as a pitch on the final day of the course. The course will be graded pass or fail. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.6 states the following: The result of a professions-act examination is valid for a two year period, until start of the clinical internships.. Participation in the Internship courses [TM20001, TM20002, TM20003, and TM20004] depends on the condition of having successfully completed TM10004, TM10006, TM10008, and TM10009 within 24 months prior to enrolment in TM20001. In the event the student does not meet this criterion, (s)he needs to resit that particular course. Upon successful completion of the resit(s), the student is eligible to enrol in TM20001 (and TM20002, TM20003, and TM20004].	
Enrolment / Application	This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

TM10009	Skills in Acute Setting	2
Responsible Instructor	Drs. I. Al Amri	
Contact Hours / Week x/x/x/x	more information in course guide (blokboek)	
Education Period	1A 1B 2A 4A 4B	
Start Education	1A 1B 2A 4A 4B	
Exam Period	Different, to be announced	
Course Language	Dutch (on request English)	
Course Contents	Medical professionals are expected to be able to manage patients with acute diseases in a protocolized fashion. For this purpose, amongst others, the ABCDE approach to the acutely ill patients, Basic Life Support (BLS) and Advanced (Trauma) Life Support (A(T)LS) have been introduced. During Medical Skills 2 (MS2), the principles behind ABCDE, BLS and A(T)LS will be repeated and students will participate in scenario training. As a part of the course and in preparation of the clinical internships, specific attention will be given to the process of feedback during learning situations and crew resource management (CRM).	
Study Goals	1. Being able to use medical knowledge in situations that demand a rapid and focused approach. 2. To understand the treatment- and communication structure that is used during emergency situations. 3. To be able to apply BLS and to participate in situations where patients are treated according to the ABCDE method or A(T)LS protcol 4. To apply self-reflection and peer-to-peer feedback in situations where medical emergencies occur.	
Study Goals continuation	A 1-6; B 2,7; C1-2; D 1-7; F 1-8	
Education Method	Online and on campus when appropriate	
Assessment	Skills will be demonstrated in a clinical scenario, reported on a clinical scenario observer form. This form exists of peer-to-peer feedback, feedback from the tutor and a self-reflection form. The combination of these three forms the report, which will be graded pass or fail. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.6 states the following: The result of a professions-act examination is valid for a two year period, until start of the clinical internships.. Participation in the Internship courses [TM20001, TM20002, TM20003, and TM20004] depends on the condition of having successfully completed TM10004, TM10006, TM10008, and TM10009 within 24 months prior to enrolment in TM20001. In the event the student does not meet this criterion, (s)he needs to resit that particular course. Upon successful completion of the resit(s), the student is eligible to enrol in TM20001 (and TM20002, TM20003, and TM20004].	
Enrolment / Application	Every 2 weeks a new group starts, both in semester 1 and 2. For each group, the course is scheduled in 5 consecutive weeks. The student has to enroll her/himself in a specific group via Brightspace after registering in MyTUDelft. Note that this course can be taken in the same semester as TM10004, but NOT in the same five week period. This course is only open to fulltime TM-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

TM Track Imaging & Intervention

TM11001	Advanced Image Acquisition	5
Responsible Instructor	Dr.ir. M. van Straten	
Course Coordinator	Dr.ir. D.H.J. Poot	
Instructor	T.P.A. O'Reilly	
Instructor	E.A.H. Warnert	
Instructor	J.A. Hernandez-Tamames	
Contact Hours / Week x/x/x/x	8/x/x/x/x/x/x/x	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	In the BSc Clinical Technology all imaging modalities have been introduced in the module Imaging bij grote ziektebeelden. In this course the physics, acquisition techniques and signal processing underlying ultrasound imaging, X-ray imaging, Computed Tomography (CT), and Magnetic Resonance Imaging (MRI) will be explored. Furthermore, the application of these imaging techniques in biomedical research and clinical practice will be discussed. The following topics will be covered: Ultrasound imaging - Physical and mathematical principles of ultrasound imaging - US probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. X-ray imaging and CT - Basic physical principles of X-ray image formation and tissue interaction - Basic principles of planar X-ray imaging and scintigraphy (short) - Advanced methods for image acquisition using CT - Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches - Issues of image quality, especially related to resolution, signal-to-noise ratio, quantitative and dynamic imaging, and dose. MRI - Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast - Spatial encoding: K-space, gradient echo, spin echo. - Advanced MR techniques such as diffusion, MR spectroscopy and perfusion. - Safety and hardware in MR imaging.	
Study Goals	- The student has in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; - The student is able to solve advanced problems addressing the theory by combining mathematical skills and physical insight; - The student is able to acquire new knowledge about clinical applications of medical imaging. - The student is able to read, discuss, summarize and comment on scientific journal and conference papers in this area. - The student can translate the theoretical knowledge of the basics of MRI into specific clinical applications	
Study Goals continuation	Contribution to learning outcomes: A1, D1, E4, F2	
Education Method	Advised - Lectures Mandatory - Working groups	
Literature and Study Materials	Books required: 9780521190657 | Webb, A. - Berrie Smith, N. | Introduction to Medical Imaging | year: 2010 Edition: 1. ook verplicht boek bij KT2301 V 9780132145183 | Price, J.L. - Links, M. | Medical Imaging Signals and Systems | year: 2014, Edition: 2 Additional study material will be provided via Brightspace	
Assessment	Written examination (90%) Assignments; pass/fail and graded (10%) The exam will contain open questions. The assignments should be done and a passing grade is required.	
Enrolment / Application	This course is only open to fulltime TM-I&I students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	Dr.ir. M. van Straten marcel.vanstraten@erasmusmc.nl Dr. D.H.J. Poot d.poot@erasmusmc.nl	

TM11002	Molecular Imaging and Therapy	5
Responsible Instructor	Dr. W. Grootjans	
Course Coordinator	Drs. L.G. Graven	
Contact Hours / Week x/x/x/x	x/8/x/x/x/x/x/x	
Education Period	1B	
Start Education	1B	
Exam Period	1B 2B	
Course Language	English	
Course Contents	In the BSc Klinische Technology, molecular imaging has already been introduced in the module Imaging bij grote ziektebeelden. In this MSc course the physics, acquisition techniques and signal processing underlying various molecular imaging techniques such as PET, SPECT, optical imaging, and spectroscopy will be explored, whereas diagnostic and therapeutic applications, both in clinical practice and preclinical research, will be outlined as well. The following topics will be covered: -Basic principles of gamma radiation, PET and SPECT, optical imaging, and spectroscopy -The indications for the different modalities -The pharmacodynamics and the development of tracers Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. Molecular imaging is a scientific field that emerged at the beginning of the twenty-first century. This field originated from the field of radiopharmacology due to the requirement for improved understanding of molecular mechanisms and pathways in organisms and cells. Therefore, the field of molecular imaging itself focuses on imaging techniques that enable non-invasive visualisation of cellular function and molecular processes, with the objective to improve the understanding of disease mechanisms, diagnosis and prognosis of disease, patient stratification, and evaluation of treatment response. Many of these techniques are now being applied in the fields of oncology, neurology and cardiovascular diseases in both a research setting and within the clinical arena. Molecular imaging differs from traditional imaging in that specific probes are used to visualize specific targets or molecular pathways. This is in contrast to conventional imaging techniques that rely on differences in physical properties (e.g. density or water content) to image the anatomy of a biological system. Imaging probes, can interact or be altered according to the state of a biological system. Detection of these probes makes it possible to either qualitatively or even quantitatively assess the state of such biological systems. Imaging probes encompass a wide range of biological agents (e.g. enzymes substrates, antibodies) and can either be administered or is endogenous to the system. Molecular imaging can be categorized in a number of different groups, depending on how a specific technique is used to assess the state of a biological system (e.g. magnetic resonance, optical detection, scintigraphy). Although many molecular imaging techniques can be used to image living cells and organisms without perturbing the biological system, other techniques such as mass spectrometry used in the operating theatre, are more destructive techniques. This course will focus on different molecular imaging techniques and their application in clinical practice as well as in preclinical research. Although many MR techniques can be categorized as molecular imaging techniques as well, the MR techniques will fall outside of the scope of this course (since these techniques have been presented and discussed in detail in the previous course, entitled advanced imaging techniques).	
Study Goals	Course learning objectives - The student has in-depth knowledge of the physics and image reconstructions underlying PET, SPECT, optical imaging and spectroscopy by combining mathematical skills and physical insight -The student is able to transfer the physical knowledge to specific patient cases and to indicate the advantages and limitations of each modality -The student is able to recognize advanced problems addressing the theory by combining mathematical skills and physical insight The student is able to acquire new knowledge about clinical applications of medical imaging -The student is able to read, discuss, summarize and comment on scientific journal papers and conference papers in this area	
Study Goals continuation	Contribution to learning outcomes: A1, D1, E5, F2	
Education Method	There will be several different forms in which the course material will be taught to the students. These include; lectures, self-study assignments, working groups, and a practical session. Attendance to the practical session is mandatory to be eligible to participate in the final exam. During the self-study hours, the students will individually read literature provided alongside the lectures and answer the questions in the course guide. These questions will test the knowledge of the student regarding these specific topics. Studying and obtaining answers to the written assignments of the self-studies in the course guide will be the responsibility of the student. The self-study assignments are made in preparation to the lectures and working groups. During the working group sessions, the students will get more practical experience regarding the topics discussed in the lectures. The students will work together in small groups to obtain answers to the working group assignments. Furthermore, the student will participate in a practical session that give hands-on experience regarding image reconstruction. Additionally, there will be a group literature assignment in which the students will work together in small groups (4-5 individuals) during the course. The exact topic of the assignment will be chosen by the students themselves, but will fall within the scope of the lectures given in this course. The assignment will serve to gain more in-depth knowledge in a particular topic and will consist of a small report and a PowerPoint presentation (the students will present their work in English). Important notice! Due to the preventive measures against the spread of the COVID19 virus, it is possible that lectures and working groups will be taught using online software tools. As of this date, it is forseen that lectures and working groups will be planned 'on campus'. However, if institutional regulations prevent 'on campus' sessions, there is room to migrate lectures and working groups to an online environment. Course participants will be informed a soon as possible on the possibilities for 'on campus' lectures and working groups.	
Literature and Study	Books required:	

Materials	9781848168565 | Rudin, M. | Molecular Imaging | year: 2013, Edition: 1 Additionals handouts wherever necessary.
Assessment	The final grade consists of: - Examination (80%) Depending on the institutional requirements to prevent spread of the COVID-19 virus, the exam can be planned 'on campus' or online using proctoring. Online proctoring is a form of online digital testing that is location-independent. Surveillance is done online, with the help of special software. For the requirements and regulations regarding online proctored exams, see https://online-learning.tudelft.nl/about/taking-exams/. - Group and/or individual assignments (20%) In this course, students will work on an assignment in groups (4-5 students) and present the outcome of the assignment. The grade of the assignment will contribute 20% to the final grade of this course.
Enrolment / Application	This course is only open to fulltime TM-I&I-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	w.grootjans@lumc.nl

TM11003	Radiation Protection	5

Responsible Instructor
Dr. K.R. Huitema

Course Coordinator
M. Schouwenburg

Contact Hours / Week x/x/x/x
x/x/8/x/x/x/x/x

Education Period
2A

Start Education
2A

Exam Period
Different, to be announced
2A
3A

Course Language
English

Course Contents
In this course radiation safety will be discussed in all its aspects. Special attention will be paid to radiation safety in hospitals. On successful completion of this course, the certificate Radiation Protection Expert (Stralingsdeskundige op het niveau van de coordinerend deskundige) will be awarded.

Short recap of content of radiation protection TMS-VRS-D (level 5A/B).
Lectures and tutorials on:
- Radiation, radioactivity, decay.
- Radiation sources.
- Interaction of radiation with matter.
- Methods of radiation detection.
- Radiation dosimetry.
- Radiation shielding.
- Biological effects of ionizing radiation.
- Internal dosimetry.
- Natural and man-made sources/background radiation.
- Radiation protection philosophy (protection principles).
- Rules and regulations; organizational, procedural and administrative tasks.
- Safety measures; operational radiation protection.
- Radiation protection when handling open sources.
- Non-ionizing radiation

Study Goals
These required (core) competencies are given by Dutch law, according to: Ministeriele regeling basisveiligheidsnormen stralingsbescherming (Rbs),
Geldend van 06-02-2018 t/m heden. Bijlage 5.2.

The three core competencies (please note; this is not an official legal translation) are:
Core competency 1: The coordinating expert is able to give, in a convincing manner, substantive advice and preventive instructions to an organisation and is able to supervise and uphold relevant law and legislation (including a granted nuclear energy act permit) in the field of ionising radiation.
Core competency 2: The coordinating expert effectively deals with an (imminent) incident or unwanted event.
Core competency 3: The coordinating expert can actively work on building his/her own competencies and those of other, mainly within the own organisation.

Study Goals continuation
Contribution to learning outcomes:
A1, D1, E5, F4

Education Method
Advised
- (Online)Lectures

Mandatory
- Practical experiments (some might be done online). These include prior preparation at home, active participation during the experiments, documenting results, reporting including discussion and conclusion/giving advice. All aspects of the practical work will be monitored and scored during the course.
- (Online) Assignments
- (Online) Special sessions: laser safety, risk perception, organisational aspects, law and regulation.

Literature and Study Materials
Books required:

Compulsory:

9789012382120 Inleiding tot de stralingsbescherming, A.A.J. Bos et al. Third print or later, can be purchased when the course starts.

or

9783527406067 | Turner, J.E. | Atoms, Radiation, and Radiation Protection. The 3rdr completely revised edition. | year: 2007, Edition: 1

Practical work book (will be made available via BrightSpace, must be printed)

Mock exams with answers (via BrightSpace)

Workbook with problems and answers (via BrightSpace)

Important websites

http://brightspace.tudelft.nl/
https://www.autoriteitnvs.nl/

BrightSpace

The student is responsible for arranging a connection with the electronic education information system BrightSpace. After logging in, the student should arrange access to the pages of the module. The student must check the latest information relating to the project on BrightSpace on a daily basis. This information is essential for good progress.

An up-to-date description of the assignments and the cases is available on BrightSpace. In addition, handouts from the lectures and additional information will be posted on BrightSpace.

Assessment
You are only allowed to take part in the exam after doing all the practical work and the assignments. The practical work is

	"valid" for a period of 2.5 years. After this period you would have to redo the practical work in order to take part in the exam. The final grade consists of: - part 1: Multiple Choice (written closed book exam), (max 33 points) - part 2: 4 problems (essay type questions, written open book exam), (max 67 points) Examinering wordt, in verband met de landelijke erkenning van het bijbehorende diploma tot stralingsbeschermingsdeskundige en in overleg met de studiecoördinator, als volgt: on campus-meerkeuze examen gevolgd door open boek examen. In het geval van onvoorziene omstandigheden of maatregelen als gevolg van bijvoorbeeld COVID-19, wordt de voorgeschreven wijze van toetsen: online tentamen, 15 korte vragen waarmee parate kennis wordt getoetst gevolgd door een open boek tentamen (2 vragen), het online tentamen zal niet leiden tot een landelijk erkend diploma tot stralingsbeschermingsdeskundige. The preferred examination method is on-campus as describe above. If this is not possible, due to COVID-19 or other measures, the exam will be replaced by an online exam. This exam will consist of 15 short ready knowledge questions, followed by 2 essay type questions (written open book). The online exam can not result in a Dutch diploma for a radiation protection expert.
Enrolment / Application	This course is only open to fulltime TM-I&I students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen Please Note: This exam is special in the sense that is a Dutch national exam, this means that the results are only available after the national exam committee has been in a meeting about the results. In general this is 8-9 weeks after the exam.
Contact	Klazien Huitema, k.r.huitema@tudelft.nl Marcel Schouwenburg, m.schouwenburg@tudelft.nl

TM11004	Radiation Therapy	5
Responsible Instructor	Dr.ir. J. Schiphof-Godart	
Course Coordinator	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	0/0/0/16/0/0/0/0 Approx. per week: - Lectures 6hrs - Presentations 2hrs - Assignments 10hrs - Selfstudy 10hrs	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Course Contents	Radiotherapy and its technologies will be explained during this course, as well as the clinical indications. The following topics will be covered in depth: Physics of ionizing radiation Technology of radiotherapy Dose calculation using analytical and Monte Carlo schemes Treatment planning: dose shaping and plan optimization Special procedures: IMRT, VMAT, proton therapy Role of radiotherapy in oncology Main clinical treatment sites: aims, approaches, outcome Clinical outcome: tumor control, risk of complication` Dose-effect modelling Radiobiology Target delineation and multi-modality imaging Research and challenges in radiotherapy Proton therapy	
Study Goals	The student has in-depth knowledge of the role of radiotherapy in treating cancer; The student has in-depth knowledge about radiobiology and the physics underlying radiotherapy; The student is able to solve advanced problems addressing the afore mentioned theory by combining mathematical skills and physics insight; The student can explain treatment planning and dose optimization and administration for various modes of radiotherapy, identify possible sources of error and apply correction methods needed for high precision radiotherapy; The student is able to read, discuss, summarize and comment on scientific journal and conference papers in this area; The student is able to present relevant literature to other students and summarize important key messages; The student is aware of daily routine in a radiotherapy department and is able to discuss relevant problems and weigh possible alternatives.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, D1, E5, F4	
Education Method	Advised - Lectures Mandatory - Working groups and group presentations - Technical assignment	
Literature and Study Materials	9789201073044 | Podgorsak, E.B. | Radiation Oncology Physics. A Handbook for Teachers and Students | Year: 2005, Edition: 1 digital available on the IAEA website.	
Assessment	The final grade consists of: - Written examination (50%) - Group assignments: technical exercises, paper and presentation, weighted with individual input. (50%) Both elements should be 5.0 or higher. Weighted average should be 5.8 or higher.	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-I&I-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	J.Schiphof-Godart@tudelft.nl	

TM11005	Advanced Image Processing	5
Responsible Instructor	Dr. J.F. Veenland	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	4A	
Start Education	4A	
Exam Period	4A 5AB	
Course Language	English	
Expected prior knowledge	The student has - Succesfully completed the Python course, or is demonstrably skilled in Python and - BSc level knowledge of image processing (level of KT3352)	
Course Contents	Advanced Medical Image Processing familiarises students with the basic methods that are used to plan and guide interventions and to extract quantitative information from images. In Medical Image Processing, segmentation, registration, classification, clustering and visualisation are crucial. For example to automatically delineate organs at risk before the planning of radiotherapy, registration and segmentation methods are applied. For the automatic quantification of the white matter lesion load in the brain, segmentation, registration, classification and visualisation are deployed. In this course the following topics will be covered: - Image segmentation - Image registration - Machine learning for images - Deep learning - Visualisation	
Study Goals	- The student has insight into state of the art algorithms for image processing including basic segmentation methods, atlas based segmentation, registration methods, feature extraction, radiomics, machine learning, deep learning and visualisation techniques. - The student is able to apply different methods to well-defined problems. - The student knows the advantages and limitations of the different methods. - The student is able to read, summarize, discuss, and comment on scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, D1, E4, F6	
Education Method	Advised - Lectures - Working groups - Self study - Programming Mandatory - Individual assignments in Python - Group assignment	
Literature and Study Materials	Study material will be listed in the course guide.	
Prerequisites	Image Processing (KT3352) Python skills (TM10007)	
Assessment	The final grade consist of: - Written examination (60%) - Assignments: reports and code of the individual assignments and group paper presentation (this is weighted with individual input) (40%) Both should be 5.0 or higher. Weighted average should be 5.8 or higher.	
Enrolment / Application	This course is only open to full time TM-I&I-students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	j.veenland@erasmusmc.nl	

TM11006	Image Guided Interventions	5
Responsible Instructor	Dr. J.S.D. Mieog	
Course Coordinator	Dr. R.W.C. Scherptong	
Course Coordinator	Dr. M.C. Burgmans	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4B	
Course Language	English	
Course Contents	Image Guided Interventions familiarizes students with the concepts of (minimally invasive) medical interventions, the pivotal role of imaging and guidance for these interventions, the clinical domains (specialities) in which these techniques are applied and the corresponding technical background. The course is structured around the following clinical domains: - Neurological interventions and aortic disease - Cardiovascular interventions - Hepatobiliary and pancreatic interventions In addition to the comprehensive elaboration on the use of imaging for interventional purposes, the students will be challenged to examine the role of technoligical developments in this field.	
Study Goals	- The student understands the principles and techniques for image guided interventions - The student has insight into their clinical applications - The student knows the advantages and limitations of the different methods - The student has a clear idea on how to improve upon an existing intervention or to develop a novel intervention - The student is able to write a translational research plan and to present an improved or novel image guided intervention	
Study Goals continuation	Contribution to learning outcomes: A1, A3, D1, E4, F6	
Education Method	Advised - Lectures Mandatory - Working groups - Group assignments - Clinical observerships - Elevator pitch presentations	
Literature and Study Materials	Book(s) required: 9780323612043 | Mauro, M. | Image-guided interventions | Year: 2014, Edition: 3 In addition to this, articles will be provided prior to the course on BrightSpace	
Assessment	- Examination multiple choice and open - Group assignments: research proposal and pitch presentation, weighted with individual input. Both have a weight of 0.5 and both should be 5.0 or higher. Weighted average should be 5.8 or higher.	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-I&I-students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	Sven Mieog, MD, PhD Department of Surgery, LUMC j.s.d.mieog@lumc.nl	

TM11007	Biomaterials and Issue Biomechanics	5
Responsible Instructor	Dr. J. Zhou	
Course Coordinator	Prof.dr. A.A. Zadpoor	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	3A	
Start Education	3A	
Exam Period	3A 4A	
Course Language	English	
Course Contents	Reconstructive surgery of bones and joints in orthopaedics, traumatology and maxillofacial surgery is mostly done using generic implants. In cases of gross bone loss, for example after resection of a bone tumour, bone resorption around a loosened prosthesis, or in cases of severe trauma, it may be impossible to adapt the bone to the prosthesis as bone stock is limited. In these cases, the implant has to be adapted to the bone and a custom made, patient specific implant has to be produced. Such an implant should not only fit the patients anatomy, but should also be mechanically stable, biocompatible, and osteoconductive. But as bone is a living tissue that is able to adapt its mass and architecture to changes in external loads, changes in the loading of the bone caused by implantation of prostheses will induce changes in bone mass. In developing implants, these changes should be predicted for an optimal long lasting result. The student will be acquainted with knowledge on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course, the students is able: To identify the type of implantable biomaterials and their main characteristics for different applications, mainly in orthopaedics, tissue engineering and regenerative medicine, and to describe the biomaterial-related aspects of implants clinical performance To explain the mechanical behaviour of biological tissues under (musculoskeletal) loading conditions and to have a basic understanding of various mechanical testing and modelling methods for tissues in the body To be familiar with available 3D printing technologies, their principles, advantages and limitations and to understand the design principles for implantable medical devices by making use of the design freedom offered by 3D printing To delineate the distinguishable aspects of regenerative medicine relative to the other therapeutic approaches to human diseases, and to define and justify the type of cells, biological factors, biomaterials and bioengineering processes used in a given pathology. To perform small-scale research in the field of biomaterials and tissue biomechanics under supervision by making use of the basic research methodology learned through performing an assignment	
Study Goals continuation	Contribution to learning outcomes: A1, D1, E4, F7	
Education Method	Mandatory - Lectures - Group assignment on selected topics	
Literature and Study Materials	Books required: 9780123746269 | Ratner, B.D. Hoffman A.S. | Biomaterial Science - An Introduction to Materials in Medicine 3rd edition | Year: 2013, Edition 3 EBOOK available in library 9781441922724 | Cowin, S.C. - Doty S.B. | Tissue Mechanics | Year: 2010, Edition: 1 EBOOK available in library 9781493944552 | Gibson, I. | Additive Manufacturing Technologies | Year: 2021, Edition: 3 EBOOK available in library Book: Tissue Engineering, C. van Blitersijk, et al., 2007 Book: Functional Metamaterials and Metadevices, Tong, Xingcun Colin, Bolingbrook, IL: Springer, 2018 (selected chapters) Book: Materials Design inspired by Nature: Function through Inner Architecture, Fratzl, Peter, John Dunlop, and Richard Weinkamer, eds., Royal Society of Chemistry, 2015 (selected chapters) Research papers related to the chosen assignment Additional material will be provided on Brightspace	
Assessment	The final grade consists of: - Written examination - Assignments The weight of the written exam is 40% and both grades should be 5.0 or higher. The period of validity is conform TER-TM article 22.	
Permitted Materials during Tests	None	
Enrolment / Application	This course is only open to fulltime TM-I&I students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

Contact	Prof.Dr. Amir Zadpoor, Tel: 81021, E-mail: A.A.Zadpoor@tudelft.nl Assoc. Prof.Dr. Jie Zhou, Tel: 85357, E-mail: J.Zhou@tudelft.nl

TM11008	Computer Assisted Reconstructive Surgery	5
Responsible Instructor	Dr.ir. B.L. Kaptein	
Course Coordinator	R.J.P. van der Wal	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	4B	
Start Education	4B	
Exam Period	4B 5AB	
Course Language	English	
Expected prior knowledge	Presumed knowledge before the course The student: Knows the principles of different medical imaging modalities such as CT, MRI and X-ray. Knows the principles of medical image processing. Knows the principles of medical image registration. Knows the anatomical planes, directions and angles. Has knowledge about Computer Aided Design (CAD). Has knowledge about surgical navigation. Has knowledge about 3D printing.	
Course Contents	Computer Assisted Reconstructive Surgery (CARS) represents a combination of computer assisted technologies that are used to improve reconstructive surgery of bones that are partially resected or deformed as a result of disease or trauma. In general these technologies are: Pre- and intra- operative (3D) imaging; pre-operative planning; patient specific implant selection (or manufacturing) and surgical navigation. By using these advanced techniques, resection of bone tumors can be done with less damage to healthy tissue and reconstruction of bones can be performed more accurately. The advanced surgical planning and navigation technology is used both pre- as well as intra operatively, and adds to the complexity of the intervention. The Technical Medicine professional will assist the surgeon in the pre-operative planning phase as well as intra-operatively by operating the (integrated) imaging and navigation system. Combining CT, MRI, and other medical image data results in an improved insight in the boundaries and location of (bone) tumors resulting in improved pre- operative planning and more accurate surgery. Based on medical image data and using additive manufacturing techniques, patient-specific implants and guiding tools can be manufactured to reconstruct the patients unique anatomy. Due to the current reliance on the expertise of the surgeon CARS is time consuming and costly and therefore only utilized in cases where patient specificity has an important added value for the clinical success of surgery that outweighs the costs. By involving the Technical Medicine professional who has the necessary medical and engineering skills in the whole CARS process, it can be optimized and streamlined to a large extent. While having the same person also present during surgery to guide surgical navigation, he or she will support the surgeon which will result in better clinical outcome for the patient. In this course the student will be trained in all aspects of computer assisted reconstructive surgery, including pre- and intra-operative imaging, pre-operative planning, implant selection and design, and per-operative navigation. Apart from the application of CARS in Orthopaedics, clinical application of CARS in other disciplines will also be part of this course.	
Study Goals	The student is able to: Describe the use of medical images in hospitals (DICOM and PACS). Describe the use of 3D visualization of medical images in hospitals. Describe the principles of SMART surgery. Describe clinical cases that cause bone defects and require patient specific instruments and implants. Describe the legal, safety and practical issues regarding implementation of patient specific implants and instruments in clinical practice. Register CT and MRI image data using (semi-)automatic image registration algorithms. Generate a 3D surface model of patient anatomy from CT and MRI image data. Design a patient specific implant and/or cutting guide that fulfills the quality requirements for 3D printing. Describe the general principles of Navigated and Robot Surgery as used within CARS. Select, justify and apply a pre-operative plan to the intra-operative situation using CARS technology.	
Study Goals continuation	Contribution to learning outcomes A1, A4, D1, E5, F8	
Education Method	Mandatory Practicals Mini symposia Individual and group assignments	
Literature and Study Materials	Compulsory 9783319129426 | Ritacco, L.E. - Milano, E. - Chao, E. | Computer-Assisted Musculoskeletal Surgery, Thinking an Executing in 3D | Year: 2015, Edition: 1 EBOOK available in library No ISBN | Bernhard Preim, Charl Botha | Visual Computing for Medicine Theory, Algorithms, and Applications, SECOND EDITION. EBOOK available in library See Course Guide	
Assessment	The final grade consists of: Written examination on campus. (50%) Assignments: Students create a portfolio containing the presentations and reports for the different assignments. (50%) Both should be 5.0 or higher. Weighted average should be 5.8 or higher. Students must have actively and sufficiently worked on the assignments to get access to the written examination. The assignment results are also valid in the academic year direct following the academic year in which the assignment results were obtained.	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-I&I students.	

	To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	Bart Kaptein M.Sc., Ph.D. B.L.Kaptein@lumc.nl 071 5264542

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

TM Track Sensing & Stimulation

TM12001	Advanced Signal Acquisition	5
Responsible Instructor	Dr.ir. W. Mugge	
Course Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	contact hours: approx 12 per week	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	The Technical Medicine professional must be able to analyze which type of measurement needs to be performed to assess physiological signals. He must perform these measurements to detect deviations from normal or certain pathologies. For this purpose, he needs fundamental knowledge on the characteristics of sensors and measuring techniques.In this course we will emphasize theoretical and practical topics regarding sensors, the acquisition of signals and the recording of those signals. The clinical application will be found in successive courses, like TM12002. The sensors that are introduced in this course can be applied to clinical measurements in audiology, cardiology, pulmonology, neurology etc. They measure physiologic parameters like blood pressure, blood flow, EMG, ECG and EEG. To maximize overall system sensitivity and accuracy students will be familiarized with the detection limit of electrical and non-electrical quantities; typical applications in the electrical, mechanical, thermal, optical and magnetic signal domain will be analyzed along with circuit and system techniques. For electrical quantities, the detection limit in a typical measurement instrument is determined by offset, finite common-mode rejection, noise and interference. The dominant source of uncertainty is identified and the equivalent input voltage/current sources are calculated. For non-electrical quantities, the detection limit should be expressed in terms of the non-electrical input parameter of interest. Important factors are: (cross-)sensitivities, non-electrical source loading and noise in the non-electrical signal domain. Furthermore, the working mechanism of the other electronic components of a measuring system like amplifiers, filters, AD and DA convertors and computer interfacing will be explained. After the course, the student should be able to design, simulate and build (simple) electronic circuits. In the end, the student will be able to discuss about the limitations and specifications of a measurement system on a professional level with an electronic expert. In collaboration with an expert, the specification of a new measurement system can be defined as a starting point for a design.	
Study Goals	1. The student is able to apply instrumentation techniques for measuring physiological signals. 2. The student is able to discuss limits of measurements techniques in relation to clinical relevance. 3. The student understands the working mechanism of the most important electronic components of a measuring system like sensors, amplifiers, filters, AD and DA convertors. 4. The student understands the most important sources of electromagnetic interference and knows how to minimize electromagnetic interference in the measured signals 5. The student understands measurement specifications, and is able to communicate on a professional level with electronics experts. 6. The student is able to define specifications of a new measurement system in collaboration with electronic experts. 7. The student is able to read, discuss, summarize and command on a scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, C2, E4, F2	
Education Method	Advised - Lectures Mandatory - Projects in groups	
Literature and Study Materials	Book(s) required: 9781292114064 | Storey, N | Electronics: a systems approach | Year: 2017, Edition: 6	
Assessment	The final grade consists of: -Final exam -Project and assignments	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-S&S students. To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. (Inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	see responsible instructor	

TM12002	Sensing of Neurophysiological Signals	5
Responsible Instructor	Dr. Z. Gao	
Course Coordinator	Dr.ir. P. Kruizinga	
Contact Hours / Week x/x/x/x	Contact hours: approx. 18 per week	
Education Period	1B	
Start Education	1B	
Exam Period	1B	
Course Language	English	
Course Contents	The overall aim of this course is to get in-depth knowledge of physics, acquisition techniques, and processing of electric signals with a focus on neuronal systems. The student learns how to record physiological signals, how to analyze and interpret them. To this end, the student acquires knowledge on neuroanatomy, neuropathology, neurophysiology, and neuropathophysiology. The student is trained to advise which type of measurement is needed to support clinical decision-making. Topics: EEG EMG BIS Evoked potentials (SSEP, BAEP, VEP, MEP) ICP NIRS INVOS	
Study Goals	1. The student has in-depth knowledge of neuroanatomy, neuropathology, neuro-physiology, and neuropathophysiology, and is able to define and analyze clinical problems in terms of anatomy and (patho)physiology 2. The student has in-depth knowledge of the physics and signal acquisition underlying EEG, EMG, BIS, SSEP, BAEP, VEP, MEP, and other monitors. 3. The student is able to solve advanced problems addressing the theory mentioned in 1 and 2 by combining mathematical skills and physical insight. 4. The student has knowledge of clinical applications of neurophysiological signal acquisition. 5. The student is able to read, discuss, summarize and comment on scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, E2, E3, E5	
Education Method	Advised - Lectures - Self study Mandatory - Workshops - OR visits - Student presentations	
Literature and Study Materials	Book(s) advised: 9781605357416 Purves, D. Neuroscience 6th edition 2018 paperback	
Assessment	The final grade consists of: - Written examination (60%) - Group assignments: paper and presentation of project, weighted with individual input. (40%)	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. (Inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	Contact: see responsible instructor	

TM12003	Electrostimulation of Neurophysiological systems	5
Responsible Instructor	Dr.ir. C.C. de Vos	
Course Coordinator	Prof.dr.ir. W.A. Serdijn	
Course Coordinator	Dr. V. Giagka	
Course Coordinator	Prof.dr. W.C. Peul	
Co-responsible for assignments	Dr.ir. C. Strydis	
Co-responsible for assignments	Dr. V. Giagka	
Co-responsible for assignments	Dr. D.G. Muratore	
Co-responsible for assignments	T.M. Lopes Marta da Costa	
Co-responsible for assignments	Prof.dr. F.J.P.M. Huygen	
Contact Hours / Week x/x/x/x	information about contact hours can be found in course guide (blokboek)	
Education Period	2A	
Start Education	2A	
Exam Period	2A 3A	
Course Language	English	
Course Contents	The Technical Medicine professional must be able, based on physiological and pathological insight, to apply electrical signals to influence organ (dys)function. To perform this, fundamental knowledge on the capability and limits of electrostimulation techniques is crucial. Furthermore, the Technical Medicine professional must be able to advise in the spectrum of clinical settings which type of stimulation technique is needed and must know its technical and clinical limits. The student is familiarized with different methods by which neurophysiological signals are actively initiated, inhibited or otherwise influenced with the goal to improve the function of the central and/or peripheral nervous system. The student is also familiarized with the physics and technical aspects of electrostimulation of visceral organs. Although the main focus of the course in on electro stimulation it also takes into account the often multi-modal and multi-disciplinary approach needed for patients who eventually need a form of neuromodulation. For a proper understanding of the effective use of electro stimulation the student also needs knowledge of (patho)physiology, the burden of the disease for the patient, the diagnostic process and alternative treatment modalities. The treatments discussed will be e.g. electroconvulsive therapy (ECT), pain modulators (e.g. SCS, DRG, TENS), deep brain stimulators, cochlear implants and transcranial magnetic stimulation (TMS).	
Study Goals	- The student has in-depth knowledge of neuroanatomy and neurophysiology - The student has in-depth knowledge of the options that are currently available for neurostimulation treatments and their mechanisms of action. - The student has in-depth knowledge of the physics and technical aspects of neurostimulation of the central and peripheral nervous system. - The student has in-depth knowledge of the physics and technical aspects of electrostimulation of visceral organs. - The student has knowledge of alternative treatment modalities, such as pharmaco- and psychotherapy. - The student is able to solve advanced problems addressing the theory mentioned above by combining mathematical skills and physical insights. - The student is able to identify risks and clinical limitations of available neurostimulation technologies. - The student is able to acquire new knowledge of clinical applications of electro-stimulation and propose new applications and treatments. - The student has in-depth knowledge of how effects of neurostimulation treatments can be assessed measured, and quantified. - The student is able to read, discuss, summarize and comment on scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, E2, E4, E5	
Education Method	Advised - Lectures Mandatory - Group assignment of project - Patient demonstration of treatment session	
Literature and Study Materials	The course material will be provided on Brightspace and during classes.	
Assessment	The final grade consists of: - Written examination - Group assignments: paper and presentation of project, weighted with individual input.	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. (Inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

Contact	Students can contact the course coordinator at: c.devos@erasmusmc.nl

TM12004	Drug Sensing and Delivery	5
Responsible Instructor	Dr. W.M.C. Mulder	
Course Coordinator	Dr. F.A.L.M. Eskens	
Contact Hours / Week x/x/x/x	an average of 8 hours per week	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Course Contents	The growth of new diagnostic methods and advances in information technology has led to the recent rise of precision and personalized medicine. Essential for successful personalized medicine is getting an optimal concentration of a drug at the precise location of action and knowing what should be measured to attain the best possible effect. Students acquire in depth knowledge in pharmacokinetics and -dynamics, drug delivery systems and controlled drug release. Students learn established and novel techniques to determine patient specific characteristics (e.g. genomics and metabolomics). Students use these characteristics to predict an individual's response and tailor the treatment. Topics: - Pharmacokinetics - Pharmacodynamics - Drug delivery systems - Controlled-drug-release - Active implantable medical devices - Closed loop systems (feedback control) - Biomarkers / metabolomics - DNA analysis / genomics - Fluid dynamics (minimizing sample size) - Follow-up and compliance	
Study Goals	Students are able to: 1. The student has knowledge about the concepts of feedback-control and can apply this knowledge for analysis of biological and design of technical systems. 2. The student has knowledge about state-of-the art diagnostic and therapeutic tools regarding drug sensing and delivery. 3. Student is able to solve advanced diagnostic and therapeutic problems regarding drug sensing and delivery, using this knowledge and tailor care to individual patients (in combination with the theory mentioned in 1). 4. The student is able to acquire new knowledge about clinical applications of feedback control systems, drug delivery systems and personalized medicine in general. 5. The student is able to read, discuss, summarize and comment on scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, E2, E3, E5	
Education Method	Advised: - Lectures, practices Mandatory - Groupassignments	
Literature and Study Materials	The course material will be provided on Brightspace	
Assessment	The method of assessment will be: - Written examination (60%) - Individual assignment (paper) (40%)	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. (Inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

TM12005	Advanced Signal Processing	5
Responsible Instructor	Prof.dr.ir. A.C. Schouten	
Course Coordinator	N.M.S. de Groot	
Contact Hours / Week x/x/x/x		
Education Period	3A	
Start Education	3A	
Exam Period	3A 4A	
Course Language	English	
Course Contents	The Technical Medicine professional must be able to analyze and interpret measurements performed on the human body. The student is familiarized with system identification techniques to understand and interpret the meaning of measurements and their relevance in the diagnosis and monitoring of a patients treatment. The students learn parameter estimation techniques to model physiology and predict response to treatment. System identification is an important tool to estimate the dynamics of a system using input and output data. It enables the characterization of physiological systems for various conditions in a standardized way, in order to gain more insight into this system. During this course the mathematical background of system identification in both time domain and frequency domain is given, including closed-loop systems (i.e. systems under feedback). Furthermore, methods will be presented to translate the identified system dynamics into physical parameters using physical models (parameter estimation). During the course examples from both technical and physiological systems will be discussed.	
Study Goals	1) The student is able to design test signals to identify unknown system dynamics, based on - signal power, sample frequency, observation time - stochastic and deterministic signals - transient and continuous signals 2) The student is able to estimate a nonparametric model of the unknown system from recorded signals, more specific - identify models in frequency domain (FRF) and time domain (IRF) - recognize and identify open-loop and closed-loop systems - estimate linear time-invariant (LTI) and recognize non-linear time-variant systems - validate the estimated models (including coherence and residual analysis) 3) The student is able to parameterise physical models from the nonparametric estimates, more specific - derive a suitable model structure based on a priori knowledge from physics - use optimisation techniques (grid search, gradient search, genetic algorithms) to parameterise the models - assess the quality of the estimated model and model parameters (Variance-Accounted-For: VAF) 4) The student is able to read, discuss, summarize and command on a scientific journal and conference papers in this area.	
Study Goals continuation	Contribution to learning outcomes: A1, A4, E4, E5	
Education Method	Advised Lectures Mandatory Assignments	
Literature and Study Materials	Selected articles and book chapters, available as e-book from the TU campus.	
Prerequisites	TM12001 (advanced signal acquisition) or equivalent course is recommended	
Assessment	The final grade consists of: Final exam Assignments (with presentations and/or reports)	
Permitted Materials during Tests	none	
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. To this end he/she acquires a (temporary) BIG registration of 9 manual medical skills which are trained in this course. In addition (inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	
Contact	See responsible instructor.	

TM12006	Extramural Sensing and Virtual Stimulation	5
Responsible Instructor	Dr.ir. J.H. de Groot	
Course Coordinator	Dr. S.K. Schiemanck	
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4B	
Course Language	English	
Course Contents	The course TM12006 Extramural Sensing and Virtual Stimulation focusses on how technological advances in medical monitoring and interventions can extend the possibilities from clinical to trans- and extramural healthcare. Especially in the field of rehabilitation medicine, in which the unmet needs of the patient are explored by using the bio-psycho-social model, development towards home-based monitoring and interventions is crucial for optimizing clinical outcome. Students are challenged to unravel the unmet needs of complex patient cases and to present a proper solution within this technological field.	
Study Goals	The student is able to: - explain the concept of E-health - explain how to extend care with respect to monitoring, data-acquisition, information and communication technology to the transmural and extramural setting - acquire new knowledge of the clinical application of new techniques and devices for assessment and treatment in the patients own living environment - translate the patients problem/medical situation into medical/technical patient- or disease-specific solutions using the bio-psycho-social model - translate the solution into requirements for extramural assessment - translate the solution into requirements for virtual interventions - read, discuss, summarize and comment on scientific journals and conference papers in this area - use constructive formative feedback in commenting on the work of peers/colleague students	
Education Method	- Blend-it lectures & working groups - Project work	
Literature and Study Materials	Course content will be provided on Brightspace.	
Assessment	Project work will be monitored and assessed by formative feedback sessions.	
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen	

TM12007	Sensing and Stimulation of Circulation and Ventilation: Acute and Chronic Care	10

Responsible Instructor	Dr. E.G. Mik
Course Coordinator	Prof.dr. J. Dankelman
Course Coordinator	Dr. M.S. Arbous
Contact Hours / Week x/x/x/x	more information about contact hours can be found in the course guide (blokboek)
Education Period	4A 4B
Start Education	4A
Exam Period	4B 5AB
Course Language	English
Course Contents	Cardiopulmonary diseases are a heavy burden to both individual patients and society. Technological advances might aid in the prevention, prediction and treatment of acute and chronic cardiac and/or pulmonary diseases. The Technical Medicine professional must be able to monitor circulatory and respiratory systems and to stimulate these in both acute and chronic settings, based on anatomical, physiological, pathological and epidemiological insight. The students acquire in depth knowledge on the characteristics and limits of cardiovascular and respiratory sensing and stimulation techniques. Furthermore, the Technical Medicine professional must be able to advise which type of sensing and/or stimulatory technique is needed in acute and chronic settings and must know its technical and clinical limits. An important focus of this course will be monitoring and support of the circulatory and ventilator system in the chronic care setting such as in the case of heart failure and Chronic Obstructive Pulmonary Disease and Asthma. Another important focus of this course will be sensing and stimulation in a patient with acute dysregulation of the circulation and ventilation (e.g. in acute care during surgery and in the ICU). The following topics will be explored: X In depth anatomy and (patho)physiology of heart, lung and circulation, building on the knowledge acquired in the BSc-KT X Epidemiology of cardiopulmonary and vascular disease X Prevention and eHealth X Early detection of cardiovascular and pulmonary disease X Monitoring and therapy in chronic care X Circulation: monitoring the complete cardiovascular status of the patient over time with established and new biosensors (e.g. blood pressure, heart rate, rhythm, cardiac output, biomarkers) X Respiration and ventilation: monitoring the respiratory and pulmonary status of the patient over time with established and new biosensors X Monitoring and therapy in acute care X Circulation: stimulation and/or support of the cardiovascular system (e.g. pacemaker, IABP, ECMO, ECLS, LVAD, RVAD) X Respiration and ventilation: stimulation and/or support of the respiratory system (e.g. mechanical ventilation, diaphragm stimulation) X Expert systems to be applied in acute and chronic care
Study Goals	1 The student has in-depth knowledge of the anatomy, physiology, pathology and pathophysiology of the circulatory and respiratory systems. Accordingly, the student can list the anatomic parts of the respiratory and circulatory system. The student is able to explain (patho) physiological changes in acute and chronic pulmonary and circulatory derangements. The student can make a differential diagnosis when a patient presents with symptoms of a failing circulatory and respiratory system and can propose diagnostic and therapeutic options, both in the chronic and acute setting. 2 The student can propose different methods of anesthesia and can describe airway, circulatory and respiratory support during anesthesia. Also, in terms of safety and quality of care, the student is able to assess critical situations, in this case, on the OR, but also applicable to other clinical situations. 3 The student can diagnose and propose therapeutic options for different forms of circulatory shock and respiratory failure (on the ICU). And the student can propose and assess different forms of measuring the circulatory and respiratory system. 4 The student has in-depth knowledge of the concepts of feedback-control and can apply this knowledge for analysis of biological systems and the design of technical systems. For example: the student can discuss ways of monitoring the circulatory system and respiratory system. Particularly, the student can discuss technological ways of measuring the circulatory and respiratory system. 5 The student is able to actively and independently acquire new knowledge of clinical methods to sense and stimulate the circulatory and ventilatory system. For example shown in the introduction of the presentation in which current knowledge has to be reviewed, summarized and reported to be able to further analyze a clinical problem. 6 The student is able to indicate risks and safety aspects of circulation and ventilation support. But is also able to address safety and quality if care in complex environments such as the OR and ICU. For example, the student is able to do a critical incident analysis. 7 The student is able to read, discuss, summarize and comment on scientific publications in this area.
Study Goals continuation	Contribution to learning outcomes: A1, A4, E2, E3, E5
Education Method	Advised - Lectures Mandatory - Working groups - Group assignment of project - Patient demonstration¡¦s with respect to circulatory and respiratory monitors and support systems (In the OR and on the ICU)
Literature and Study Materials	The course material will be provided on Brightspace
Assessment	The final grade consists of: - Examination (70%) - Final presentation (30%) The examination and final presentation are both awarded a mark; the poster assignments receive a pass or fail and have to be substituted by an additional poster assignment in case of a fail. The final grade is

	calculated by averaging the mark for the assignment and the examination. If one of these is a fail, it is possible to revise the assignment or take an additional examination. If both of these are fails, it is not possible to revise the assignment or take an additional examination.
Permitted Materials during Tests	none
Enrolment / Application	This course is only open to fulltime TM-S&S students To participate in this course all students must register in MyTUDelft during the registration period. Without official registration in MyTUDelft, students cannot participate in the course. More information about registration for courses, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken Registration for a course does not count as registration for an examination. Students must register for examinations separately in accordance with the relevant provisions. More information about registration for exams, the procedure and relevant deadlines can be found on: https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.
Remarks	The Technical Medicine professional must be able to take full responsibility for his/her part of the patient diagnostic and therapeutic process. To this end he/she acquires a (temporary) BIG registration of 9 manual medical skills which are trained in this course. In addition (inter)professional written communication between caretakers is practiced. This is an absolute necessity for a TM professional to blend in the healthcare system.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

TM Second Year	
Contact for students	MSc-programme: master-tm@tudelft.nl TM Internships: stage-tm@tudelft.nl
In association with the University of	Leiden Universty (LUMC) Delft University of Technology (3mE) Erasmus University Rotterdam (Erasmus MC)
Prerequisites	Entry requirements apply. Check the Teaching and Education Regulations via: http://onderwijs.3me.tudelft.nl/reglementen (in the appendix)
Introduction 1	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen

TM20005	TM Clinical Internship 1	14

Responsible Instructor M. Walraven

Contact Hours / Week x/x/x/x Fulltime

Education Period
1
2
3

Start Education
1A
2A
3A

Exam Period Different, to be announced

Course Language Dutch

Course Contents

During the second year of the Master Technical Medicine (TM) students are undertaking their first steps in the work field to experience their own role in a clinical work environment. This happens through four so called Clinical Internships The first aim of these internships serves to obtain the clinically required skills. Secondly the student works on a medical technical question, from this research multiple products will result (e.g. a report, technology, or knowledge). The third aspect is the reflection on the student's own identity considering responsibility towards patients and other actors in healthcare.

The clinical internships have learning objectives based on four components: Technical Medicine mentality, medical functioning, technical functioning, professional behaviour.

In the 10 week period of the internship the internship takes place from Monday to Thursday in a clinical institution. In every uneven week there is education and group meetings to discuss professional and personal development. The Fridays in even weeks are meant to work on the report or assignments for the internship.

During the year 2023-2024 the internships are planned in the following periods:
- Q1: 11 Sept - 17 Nov
- Q2: 4 Dec - 23 Feb (2 weeks off during the holidays)
- Q3: 11 March- 17 May
- Q4: 3 June- 9 Aug

Please note internship 1 is only available in Q1, Q2 and Q3. This means you cannot start your first internship during the summer period (June-Aug).

Study Goals

Learning objectives Clinical Internship:

1.Technical Medicine (TM) component
-The student is able to translate a medical problem into a TM project with a proper aim and project plan.
-The student is able to function as a starting professional with enough (technical) knowledge to have a substantive conversation about the TM project.
-The student has (developed) a personal and recognizable TM mentality and working method which is notable during conversations and in output.
-The final internship product (report/presentation) meets scientific stadards both in structure as well as in lay-out and style.

2. Medical component
- See the learning objectives in Dutch

3. Technical component
- The student has achieved the (technical) aims from the project plan.
- The student has gained insights from the scientific and technical domain to analyze the TM problem. The student uses the gained knowledge to substantiate the project plan.
- The student understands and employs academic reasoning and the processes around (technical) innovations.
- The student analyses and justifies his/her chosen methods in a clear and understandable way, both in the project plan and at the end of the internship.
- The student makes all required results of the project available in a structured and understandable way, i.e. protocols, data, analyses, specifications and manuals.

4. Professional behavior
-The outcome of the project is the result of the students own initiative, insight and implementation.
-The student shows that he/she has the capability to solve problems and adjust the project plan based on findings from literature or practical experience.
-The student asks for feedback and is able to receive and use given feedback. The student has an open and reflective attitude.
-The students behavior towards colleagues and other disciplines is respectful and professional.
-The student is present at the department in a pleasant and professional way.

Education Method

4 days internship and 1 day group education or independent study
- Fridays of uneven internship-weeks (1, 3, 5, 7, 9) education and intervision
- Fridays of even weeks (2, 4, 6, 8, 10) work on internship report or internship-assignments

Prerequisites

For the first Clinical Internship: 45 EC need to be obtained in the first master year, including: All 5 patient cases, both medical skills courses and the patient journey course. The director of studies can give exemptions on these prerequisites.
For the second internship: 60 EC need to be obtained in the first master year.
For the third internship: 60 EC need to be obtained in the first master year and 15 EC need to be obtained in the second year.
For the fourth internship: 60 EC need to be obtained in the first master year and 30 EC need to be obtained in the second year. The director of studies can give exemptions on these prerequisites.

Please note: Internship 1 is only available in Q1, Q2 and Q3 (start in Sept, Dec, March). Internship 1 cannot take place in Q4 (start in June).

Assessment

1. Sort of assesments
-Week 2: Plan of approach(Go / No-go)
-Week 6: Midterm evaluation (Formative)
-Week 10: Final evaluation by medical and technical supervisor *
-Week 10: Reflection by "intervisie-docent"
-Week 1-10: Clinical practical observations** (Formative)

2. Calculation of final grade:
- Pass from medical supervisor
- Pass from technical supervisor
- Pass from "intervisie-docent"
- Required minimum of 6 clinical practical observations

3. Additional information
* The final evaluation by the supervisors is based on the learning goals for this internship. Formative feedback is given by all supervisors. All supervisors need to give a "pass" to successfully conclude the clinical internship.

	** The clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For every TM Clinical Internship an average of 10 observations (signed forms) is asked with a required minimum of 6 observations and a maximum of 14 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. To participate in this course register on Brightspace for the 'TM-internship'-page or send an email to stage-TM.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen
Contact	stage-tm@tudelft.nl

TM20006	TM Clinical Internship 2	14
Responsible Instructor	M. Walraven	
Contact Hours / Week x/x/x/x	Fulltime	
Education Period	1 2 3 4	
Start Education	1A 2A 3A 4A	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	During the second year of the Master Technical Medicine (TM) students are undertaking their first steps in the work field to experience their own role in a clinical work environment. This happens through four so called Clinical Internships The first aim of these internships serves to obtain the clinically required skills. Secondly the student works on a medical technical question, from this research multiple products will result (e.g. a report, technology, or knowledge). The third aspect is the reflection on the student's own identity considering responsibility towards patients and other actors in healthcare. The clinical internships have learning objectives based on four components: Technical Medicine mentality, medical functioning, technical functioning, professional behaviour. In the 10 week period of the internship the internship takes place from Monday to Thursday in a clinical institution. In every uneven week there is education and group meetings to discuss professional and personal development. The Fridays in even weeks are meant to work on the report or assignments for the internship. During the year 2023-2024 the internships are planned in the following periods: - Q1: 11 Sept - 17 Nov - Q2: 4 Dec - 23 Feb (2 weeks off during the holidays) - Q3: 11 March- 17 May - Q4: 3 June- 9 Aug Please note internship 1 is only available in Q1, Q2 and Q3. This means you cannot start your first internship during the summer period (June-Aug).	
Study Goals	Learning objectives Clinical Internship: 1.Technical Medicine (TM) component -The student is able to translate a medical problem into a TM project with a proper aim and project plan. -The student is able to function as a starting professional with enough (technical) knowledge to have a substantive conversation about the TM project. -The student has (developed) a personal and recognizable TM mentality and working method which is notable during conversations and in output. -The final internship product (report/presentation) meets scientific stadards both in structure as well as in lay-out and style. 2. Medical component - See the learning objectives in Dutch 3. Technical component - The student has achieved the (technical) aims from the project plan. - The student has gained insights from the scientific and technical domain to analyze the TM problem. The student uses the gained knowledge to substantiate the project plan. - The student understands and employs academic reasoning and the processes around (technical) innovations. - The student analyses and justifies his/her chosen methods in a clear and understandable way, both in the project plan and at the end of the internship. - The student makes all required results of the project available in a structured and understandable way, i.e. protocols, data, analyses, specifications and manuals. 4. Professional behavior -The outcome of the project is the result of the students own initiative, insight and implementation. -The student shows that he/she has the capability to solve problems and adjust the project plan based on findings from literature or practical experience. -The student asks for feedback and is able to receive and use given feedback. The student has an open and reflective attitude. -The students behavior towards colleagues and other disciplines is respectful and professional. -The student is present at the department in a pleasant and professional way.	
Education Method	4 days internship and 1 day group education or independent study - Fridays of uneven internship-weeks (1, 3, 5, 7, 9) education and intervision - Fridays of even weeks (2, 4, 6, 8, 10) work on internship report or internship-assignments	
Prerequisites	For the first clinical internship: 45 EC need to be obtained in the first master year, including: All 5 patient cases, both medical skills courses and the patient journey course. The director of studies can give exemptions on these prerequisites. For the second internship: 60 EC need to be obtained in the first master year. For the third internship: 60 EC need to be obtained in the first master year and 15 EC need to be obtained in the second year. For the fourth internship: 60 EC need to be obtained in the first master year and 30 EC need to be obtained in the second year. The director of studies can give exemptions on these prerequisites. Please note: Internship 1 is only available in Q1, Q2 and Q3 (start in Sept, Dec, March). Internship 1 cannot take place in Q4 (start in June).	
Assessment	1. Sort of assesments -Week 2: Plan of approach(Go / No-go) -Week 6: Midterm evaluation (Formative) -Week 10: Final evaluation by medical and technical supervisor * -Week 10: Reflection by "intervisie-docent"* -Week 1-10: Clinical practical observations** (Formative) 2. Calculation of final grade: - Pass from medical supervisor - Pass from technical supervisor - Pass from "intervisie-docent" - Required minimum of 6 clinical practical observations 3. Additional information	

	* The final evaluation by the supervisors is based on the learning goals for this internship. Formative feedback is given by all supervisors. All supervisors need to give a "pass" to successfully conclude the clinical internship. ** The clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For every TM Clinical Internship an average of 10 observations (signed forms) is asked with a required minimum of 6 observations and a maximum of 14 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. To participate in this course register on Brightspace for the 'TM-internship'-page or send an email to stage-TM.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen
Contact	stage-tm@tudelft.nl stage-tm@tudelft.nl

TM20007	TM Clinical Internship 3	14

Responsible Instructor	M. Walraven
Contact Hours / Week x/x/x/x	Fulltime
Education Period	1 2 3 4
Start Education	1A 2A 3A 4A
Exam Period	Different, to be announced
Course Language	Dutch
Course Contents	During the second year of the Master Technical Medicine (TM) students are undertaking their first steps in the work field to experience their own role in a clinical work environment. This happens through four so called Clinical Internships The first aim of these internships serves to obtain the clinically required skills. Secondly the student works on a medical technical question, from this research multiple products will result (e.g. a report, technology, or knowledge). The third aspect is the reflection on the student's own identity considering responsibility towards patients and other actors in healthcare. The clinical internships have learning objectives based on four components: Technical Medicine mentality, medical functioning, technical functioning, professional behaviour. In the 10 week period of the internship the internship takes place from Monday to Thursday in a clinical institution. In every uneven week there is education and group meetings to discuss professional and personal development. The Fridays in even weeks are meant to work on the report or assignments for the internship. During the year 2023-2024 the internships are planned in the following periods: - Q1: 11 Sept - 17 Nov - Q2: 4 Dec - 23 Feb (2 weeks off during the holidays) - Q3: 11 March- 17 May - Q4: 3 June- 9 Aug Please note internship 1 is only available in Q1, Q2 and Q3. This means you cannot start your first internship during the summer period (June-Aug).
Study Goals	Learning objectives Clinical Internship: 1.Technical Medicine (TM) component -The student is able to translate a medical problem into a TM project with a proper aim and project plan. -The student is able to function as a starting professional with enough (technical) knowledge to have a substantive conversation about the TM project. -The student has (developed) a personal and recognizable TM mentality and working method which is notable during conversations and in output. -The final internship product (report/presentation) meets scientific stadards both in structure as well as in lay-out and style. 2. Medical component - See the learning objectives in Dutch 3. Technical component - The student has achieved the (technical) aims from the project plan. - The student has gained insights from the scientific and technical domain to analyze the TM problem. The student uses the gained knowledge to substantiate the project plan. - The student understands and employs academic reasoning and the processes around (technical) innovations. - The student analyses and justifies his/her chosen methods in a clear and understandable way, both in the project plan and at the end of the internship. - The student makes all required results of the project available in a structured and understandable way, i.e. protocols, data, analyses, specifications and manuals. 4. Professional behavior -The outcome of the project is the result of the students own initiative, insight and implementation. -The student shows that he/she has the capability to solve problems and adjust the project plan based on findings from literature or practical experience. -The student asks for feedback and is able to receive and use given feedback. The student has an open and reflective attitude. -The students behavior towards colleagues and other disciplines is respectful and professional. -The student is present at the department in a pleasant and professional way.
Education Method	4 days internship and 1 day group education or independent study - Fridays of uneven internship-weeks (1, 3, 5, 7, 9) education and intervision - Fridays of even weeks (2, 4, 6, 8, 10) work on internship report or internship-assignments
Prerequisites	For the first clinical internship: 45 EC need to be obtained in the first master year, including: All 5 patient cases, both medical skills courses and the patient journey course. The director of studies can give exemptions on these prerequisites. For the second internship: 60 EC need to be obtained in the first master year. For the third internship: 60 EC need to be obtained in the first master year and 15 EC need to be obtained in the second year. For the fourth internship: 60 EC need to be obtained in the first master year and 30 EC need to be obtained in the second year. The director of studies can give exemptions on these prerequisites. Please note: Internship 1 is only available in Q1, Q2 and Q3 (start in Sept, Dec, March). Internship 1 cannot take place in Q4 (start in June).
Assessment	1. Sort of assesments -Week 2: Plan of approach(Go / No-go) -Week 6: Midterm evaluation (Formative) -Week 10: Final evaluation by medical and technical supervisor * -Week 10: Reflection by "intervisie-docent"* -Week 1-10: Clinical practical observations** (Formative) 2. Calculation of final grade: - Pass from medical supervisor - Pass from technical supervisor - Pass from "intervisie-docent" - Required minimum of 6 clinical practical observations 3. Additional information

	* The final evaluation by the supervisors is based on the learning goals for this internship. Formative feedback is given by all supervisors. All supervisors need to give a "pass" to successfully conclude the clinical internship. ** The clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For every TM Clinical Internship an average of 10 observations (signed forms) is asked with a required minimum of 6 observations and a maximum of 14 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. To participate in this course register on Brightspace for the 'TM-internship'-page or send an email to stage-TM.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	stage-tm@tudelft.nl stage-tm@tudelft.nl

TM20008	TM Clinical Internship 4	14
Responsible Instructor	M. Walraven	
Contact Hours / Week x/x/x/x	Fulltime	
Education Period	1 2 3 4	
Start Education	1A 2A 3A 4A	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	During the second year of the Master Technical Medicine (TM) students are undertaking their first steps in the work field to experience their own role in a clinical work environment. This happens through four so called Clinical Internships The first aim of these internships serves to obtain the clinically required skills. Secondly the student works on a medical technical question, from this research multiple products will result (e.g. a report, technology, or knowledge). The third aspect is the reflection on the student's own identity considering responsibility towards patients and other actors in healthcare. The clinical internships have learning objectives based on four components: Technical Medicine mentality, medical functioning, technical functioning, professional behaviour. In the 10 week period of the internship the internship takes place from Monday to Thursday in a clinical institution. In every uneven week there is education and group meetings to discuss professional and personal development. The Fridays in even weeks are meant to work on the report or assignments for the internship. During the year 2023-2024 the internships are planned in the following periods: - Q1: 11 Sept - 17 Nov - Q2: 4 Dec - 23 Feb (2 weeks off during the holidays) - Q3: 11 March- 17 May - Q4: 3 June- 9 Aug Please note internship 1 is only available in Q1, Q2 and Q3. This means you cannot start your first internship during the summer period (June-Aug).	
Study Goals	Learning objectives Clinical Internship: 1.Technical Medicine (TM) component -The student is able to translate a medical problem into a TM project with a proper aim and project plan. -The student is able to function as a starting professional with enough (technical) knowledge to have a substantive conversation about the TM project. -The student has (developed) a personal and recognizable TM mentality and working method which is notable during conversations and in output. -The final internship product (report/presentation) meets scientific stadards both in structure as well as in lay-out and style. 2. Medical component - See the learning objectives in Dutch 3. Technical component - The student has achieved the (technical) aims from the project plan. - The student has gained insights from the scientific and technical domain to analyze the TM problem. The student uses the gained knowledge to substantiate the project plan. - The student understands and employs academic reasoning and the processes around (technical) innovations. - The student analyses and justifies his/her chosen methods in a clear and understandable way, both in the project plan and at the end of the internship. - The student makes all required results of the project available in a structured and understandable way, i.e. protocols, data, analyses, specifications and manuals. 4. Professional behavior -The outcome of the project is the result of the students own initiative, insight and implementation. -The student shows that he/she has the capability to solve problems and adjust the project plan based on findings from literature or practical experience. -The student asks for feedback and is able to receive and use given feedback. The student has an open and reflective attitude. -The students behavior towards colleagues and other disciplines is respectful and professional. -The student is present at the department in a pleasant and professional way.	
Education Method	4 days internship and 1 day group education or independent study - Fridays of uneven internship-weeks (1, 3, 5, 7, 9) education and intervision - Fridays of even weeks (2, 4, 6, 8, 10) work on internship report or internship-assignments	
Prerequisites	For the first clinical internship: 45 EC need to be obtained in the first master year, including: All 5 patient cases, both medical skills courses and the patient journey course. The director of studies can give exemptions on these prerequisites. For the second internship: 60 EC need to be obtained in the first master year. For the third internship: 60 EC need to be obtained in the first master year and 15 EC need to be obtained in the second year. For the fourth internship: 60 EC need to be obtained in the first master year and 30 EC need to be obtained in the second year. The director of studies can give exemptions on these prerequisites. Please note: Internship 1 is only available in Q1, Q2 and Q3 (start in Sept, Dec, March). Internship 1 cannot take place in Q4 (start in June).	
Assessment	1. Sort of assesments -Week 2: Plan of approach(Go / No-go) -Week 6: Midterm evaluation (Formative) -Week 10: Final evaluation by medical and technical supervisor * -Week 10: Reflection by "intervisie-docent"* -Week 1-10: Clinical practical observations** (Formative) 2. Calculation of final grade: - Pass from medical supervisor - Pass from technical supervisor - Pass from "intervisie-docent" - Required minimum of 6 clinical practical observations 3. Additional information	

	* The final evaluation by the supervisors is based on the learning goals for this internship. Formative feedback is given by all supervisors. All supervisors need to give a "pass" to successfully conclude the clinical internship. ** The clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For every TM Clinical Internship an average of 10 observations (signed forms) is asked with a required minimum of 6 observations and a maximum of 14 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total. Period of validity: The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. To participate in this course register on Brightspace for the 'TM-internship'-page or send an email to stage-TM.
Department	MSc Technical Medicine (Joint-degree)
Judgement	All rules and regulations that apply can be found in the related documents via, http://onderwijs.3me.tudelft.nl/reglementen
Contact	stage-tm@tudelft.nl stage-tm@tudelft.nl

TM20009	**Medical Technology in Clinical Practice**	**4**
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Responsible Instructor	Dr. J.J. Baalbergen	
Contact Hours / Week x/x/x/x	Fulltime	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	De levenscyclus voor Medische Technologie is het handvat voor het centrale leerdoel: het kunnen borgen van het veilig en doelmatig gebruik van MT in de kliniek. Gedurende het studiejaar heeft ieder blok / kwartaal een thema: A. Prospectief, het veilig introduceren van medische technologie in de kliniek (najaar) B. Retrospectief, het veilig gebruiken van medische technologie in de kliniek (winter) C. Veiligheid binnen het ziekenhuis (voorjaar) D. Het sluiten van de cirkel en reflectie in het vierde kwartaal (zomer)	
Study Goals	Het kunnen borgen van het veilig en doelmatig gebruik van medische technologie (MT) in de kliniek De toepassing van MT: het gebruik van MT in de zorg en de kliniek in het bijzonder Kennis van wet en regelgeving m.b.t. MT Het herkennen van risicos m.b.t. het gebruik van MT Het kwantitatief vaststellen van doelmatig en veilig gebruik van MT Het ontwerpen (en eventueel implementeren) van verbeteringen Het vaststellen van kosten (cost of ownership) en baten, effectiviteit (wat zijn de medische effecten, vergelijken van verschillende technieken, etc.)	
Education Method	Het vak Medical Technology in Clinical Practice (MTK) loopt parallel aan de vier TM2-stages. De leerstof wordt aangeboden d.m.v. hoorcolleges en werkgroepen. Er wordt gewerkt met telkens een andere gastspreker die een bepaald onderwerp inleidt (maximaal 2 uur). De studenten verwerven kennis over het onderwerp en verwerken deze kennis in een groepssessie/opdracht. In het vierde kwartaal (in de zomer) vindt éénmaal MTK onderwijs plaats. Er dient een grotere opdracht ingeleverd te worden, het onderwerp wordt in overleg bepaald.	
Assessment	De eindbeoordeling van het MTK onderwijs is voldoende of onvoldoende en wordt per blok vastgesteld. Tijdens een blok worden 2 of 3 opdrachten aan de studenten uitgereikt. De opdrachten zullen zo veel mogelijk gelinkt zijn aan het onderwerp van de stage. In het vierde kwartaal wordt alleen één grotere opdracht uitgereikt, het onderwerp is gelinkt aan één van de kwartaalthemas. Het MTK onderwijs wordt voldoende beoordeeld indien aan de volgende voorwaarden voldaan is: Alle opgaven zijn ingeleverd en met een voldoende beoordeeld Aanwezigheid tijdens het MTK onderwijs in dit blok 80 % of hoger	
Enrolment / Application	This course is only open to full time Technical Medicine students.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen	
Contact	zie studiegids	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Technical Medicine**

TM Third Year

TM30001	TM Practical Internship	15
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Course Coordinator	M. Walraven	
Contact Hours / Week x/x/x/x	Fulltime	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	The third year of the Master Technical Medicine (TM) is the graduation year with the MSc thesis as major project. Before starting this final project students will have the option to do a Practical internship of 15 EC. The main aim of this internship is to give students the option to experience a different work environment of choice, for example to work at a company, to work/study abroad or to gain clinical experience at a healthcare institution. To facilitate optimal freedom of choice this internship has only a few requirements. - The internship is at MSc level. - The internship is linked to Technical Medicine. - The internship fits into the students personal development goals. To verify these criteria the internship has to be approved by supervisors and the mastercoordinator in your digital portfolio (Scorion).	
Study Goals	For this internship the student has to define personal study goals. These goals have to follow-up on the personal education plan (POP) the student has worked on during the TM2 year. The student is expected to reflect on the study goals at the end of the internship in a reflective report.	
Education Method	To be determined by the student in consultation with supervisors.	
Prerequisites	TM1 (60 EC) fully obtained, 3 clinical internships (45 EC of TM2) obtained and an approved graduation plan or 3ME application form. Please note that there is a possibility to start a practical internship before starting a 4th Clinical Internship (TM20008). While planning the TM3 year please note that there is a period of validity for the TM clinical internships. The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.	
Assessment	1. Assessment By the end of the internship you are required to have a document that shows your work (MSc-level) and a reflective report about your experiences. You will not receive a grade. Pass/Fail based on assessment of local supervisor, product and reflective report. 2. Grade The final grade is a PASS/FAIL based on: - Assessment local supervisor (pass); - Final product (MSc-level); - Reflective report; - Assessment of LDE-supervisor (pass) 3. Additional - Before starting an internship form (stageovereenkomst UNL) has to be approved by the supervisor and the mastercoordinator.	
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. You can register for this course in Scorion (the digital portfolio) by sending the internship contract (stageovereenkomst UNL) to stage-TM. More information can be found at Brightspace TM30001.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen	
Contact	stage-tm@tudelft.nl	

TM30002	TM Thesis Feasibility Study	15
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Course Coordinator	M. Walraven	
Contact Hours / Week x/x/x/x	various	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The third year of the Master Technical Medicine (TM) is the graduation year with the MSc thesis as major project of 45 EC. Students have the option to do a Thesis feasibility study of 15 EC before starting, during, or after finishing the MSc thesis. The aim of this Thesis feasibility study is to give students the option to extend the graduation project for example with a valorization study or by mastering skills required for the graduation project.	
Study Goals	For the feasibility study the student has to define study goals. These goals have to match with the MSc thesis and the personal education plan (POP)	
Education Method	1. Assessment To be determined by the student in consultation with the graduation supervisor. By the end of the internship you are required to have a document that shows your work (MSc-level) and a reflective report. Pass/Fail based on assessment of graduation supervisor, report and reflective report. 2. Grade The final grade is a PASS/FAIL based on: - Assessment supervisor (pass); - Final document (MSc-level); - Reflective report; 3. Additional - The thesis feasibility study has to be approved by the mastercoordinator in the graduation plan. - The thesis feasibility study can be done before, during or after the literature study (TM30003) and/or Master thesis (TM30004).	
Prerequisites	TM1 (60 EC) fully obtained, TM2 (60 EC) fully obtained and an approved graduation plan. While planning the TM3 year please note that there is a period of validity for the TM clinical internships. The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.	
Assessment	Before the start of the thesis feasibility study the graduation plan has to be approved. Send the graduation plan to your supervisor and the mastercoordinator in your digital portfolio (Scorion). By the end of the thesis feasibility study you are required to have a document/product that shows the work you have done and a reflective report. Your graduation supervisor will score your work with a pass/fail.	
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. You can register for this course in Scorion (the digital portfolio) by sending your graduation plan to stage-TM. More information can be found at Brightspace TM30002.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen	
Contact	stage-tm@tudelft.nl	

TM30003	TM Literature Study	10
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Course Coordinator	M. Walraven	
Contact Hours / Week x/x/x/x	various	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The third year of the master Technical Medicine (TM) is the graduation year with the Master thesis as major project. This project is divided in a MSc thesis part of 35 EC and a literature study of 10 EC. The MSc thesis is supervised by one medical and one technical supervisor of which one functions as graduation supervisor. The goal of the literature study is to prepare for your MSc Thesis. The reasons to start with a literature study is to quickly achieve an overview of your research topic. Apart from writing a report, you will give a presentation about the acquired overview and its implications for your master thesis.	
Study Goals	The Learning Objectives The student is able to: - Search, select, evaluate, and synthesize representative scientific sources for the topic from several perspectives (for example, economic, ethical- environmental, -and health) relevant to the topic; - apply best practices for conducting methodological searches in the literature review; - write a comprehensive and balanced, opinionated literature review that deeply explores the issues in the area of study, leading to new insights in academic language; - clearly define the purpose and objectives of the literature review; - draw conclusions related to the literature research problem and give recommendations towards new research opportunities, applications and consequences for the field; - argument a statement using the information from literature, including counter arguments. - manage the individual learning process, including time management and adequate planning (minimally exceeding allotted time);	
Education Method	The Literature study is individual work. The student and the graduation supervisor together decide on the topic. The student is in charge of the planning. Although you are in charge, this doesn't mean that you have to do this in isolation. Ask your graduation supervisor and/or your medical and technical supervisors for advice, but also contact a librarian or an expert in the field for advice on your search for literature. Writing on a scientific level is difficult, so the feedback that you receive on the drafts of your literature report will be very valuable.	
Prerequisites	TM1 (60 ECTS) and TM2 (60 ECTS) fully obtained and an approved graduation plan. While planning the TM3 year please note that there is a period of validity for the TM clinical internships. The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.	
Assessment	1. Assesment By the end of the literature study you are required to write a scientific report in academic English about your literature study. Apart from writing a report, you will give a presentation (in Dutch or English) about the overview acquired and its implications for your master thesis. 2. Final grade - 40% final grade from medical supervisor for report - 40% final grade from technical supervisor for report - 10% final grade from an independent assessor (member of the department staff) for the presentation - 10% final grade from another independent assessor (member of the department staff) for the presentation 3. Further information Before the start of the literature study the graduation plan has to be approved.	
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. You can register for this course in Scorion (the digital portfolio) by sending your graduation plan to your graduation supervisor and stage-TM. More information can be found at Brightspace TM30003.	
Department	MSc Technical Medicine (Joint-degree)	
Judgement	All rules and regulations that apply can be found in the related documents via: http://onderwijs.3me.tudelft.nl/reglementen	
Contact	stage-tm@tudelft.nl	

TM30004	TM MSc Thesis	35
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Course Coordinator	M. Walraven	
Contact Hours / Week x/x/x/x	various	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The third year of the master Technical Medicine (TM) is the graduation year with the MSc thesis as major project. This project is divided in a MSc thesis of 35 EC and a literature study of 10 EC. The MSc thesis is supervised by one medical and one technical supervisor of which one functions as graduation supervisor.	
Study Goals	The overall objective of the Master Thesis is to demonstrate a sufficient academic level in the field of Technical Medicine. The goal of the master programme in Technical Medicine is to educate medical professionals, who are technically high-skilled. The graduate meets the following qualifications to a sufficient level: - Be able to see and screen specific patient populations - Do relevant measurements, analyze and interpret results - Develop a treatment plan within the technical-medicine domain - Treat patients with medical technological interventions - Implement and evaluate new technologies in the clinical domain - Evaluate the use of specific technologies on a patient - Do academic research	
Education Method	The education method is mainly learning by doing. Together with your supervisors you are in charge of your thesis project and act as the project leader of your thesis work. This implies that you should operate on a highly independent level on one side, but also make use of all available resources on the other side. One very important resource is your supervisors who will advise you on matters that concern the thesis content, but also the process of your thesis work. It is mandatory to have regular meetings with your supervisors in which you discuss your planning, approach and all other content. In these meetings, you should make clear agreements on the goals, go-no-go moments, and deadlines of your project. It is your responsibility to ask for proper feedback and make sure that you are right on track. The exact details on how the organization of your graduation trajectory is given form, is dependent on you and your supervisors' working style. However, there are two obligatory feedback moments: in week 2 there is a go/no-go moment where your supervisors approve your work plan; in approximately week 12 (half-way) there is a formative mid-way feedback moment. To monitor clinical exposure during the master thesis project clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For the master thesis an average of 20 observations (signed forms) is asked with a required minimum of 10 observations and a maximum of 28 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total.	
Prerequisites	TM1 (60 EC) and TM2 (60 EC) fully obtained, literature study (TM30003, 10 EC) obtained and an approved graduation plan. To monitor clinical exposure during the master clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. Before starting the master thesis a student needs to have collected at least 32 observations (forms) during year 2. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 forms in total. While planning the TM3 year please note that there is a period of validity for the TM clinical internships. The Teaching and Examination Regulations (TER-TM), article 22.5 states the following: The result of a TM internship is valid for a three year period.	
Assessment	1. Assesment Students prepare the MSc thesis as a scientific paper written in academic English with complementing appendices to cover all work. The thesis project is evaluated on four aspects: the content and process of the thesis work, the scientific paper with complementing appendices, an oral presentation (graduation colloquium) by the MSc candidate and an oral examination (in Dutch or English) before an MSc examination committee. 2. Final grade Based on the grade given by the "afstudeercommissie". 3. Further information - Before the start of the MSc thesis your graduation plan needs to be approved. - During your project there will be several informal feedback sessions (week 2 and week 12). - To monitor clinical exposure during the master thesis project clinical practical observations are collected with two forms: 1) korte klinische beoordelingen (KKB) en 2) OSAT-formulieren. For the master thesis an average of 20 observations (signed forms) is asked with a required minimum of 10 observations and a maximum of 28 observations. During year 2 and 3 of the master Technical Medicine there is a requirement of 60 clinical practical observations in total.	
Enrolment / Application	This course is only open to fulltime Technical Medicine (TM)-students. You can register for this course in Scorion (the digital portfolio) by sending your graduation plan to your graduation supervisor and stage-TM. More information can be found at Brightspace TM30004.	
Department	MSc Technical Medicine (Joint-degree)	

15 EC Elective courses

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N.M.S. de Groot

Department	Signal Processing Systems

Unit	Mech, Maritime & Materials Eng
Department	Biomechanical Engineering

Dr. W. Grootjans

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Department	Biomechanical Engineering

J.A. Hernandez-Tamames

Department	ImPhys/Vos group

Prof.dr. M.S. Hoogeman

Department	RST/Medical Physics & Technology

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